AI-Based Smart Classroom Digital Twin for CO■ Monitoring and Ventilation Optimization

Final Project Technical Report

Abstract

This project builds an educational digital twin of a ventilated classroom to simulate indoor CO■ accumulation and evaluate ventilation control. Occupancy changes through a scheduled teaching day, a sensor-like reading includes random noise, and a rule controller changes ventilation based on measured concentration. The controller is compared with low and high fixed-flow baselines. The notebook produces time-series figures, a policy scorecard, a parameter sensitivity demonstration, anomaly flags and automated tests. The model is intentionally transparent and reproducible. Its inputs are illustrative, not measurements from Veltech, so results demonstrate methodology rather than real campus air quality.

1. Introduction

Classrooms are dynamic environments: occupants generate CO■ and ventilation dilutes it. A digital twin can represent the system and compare control policies before changes are made to a real building. The project question is whether sensor-driven ventilation reduces simulated CO■ accumulation compared with low fixed ventilation, and how this compares with continuous high ventilation. The model is a teaching prototype, not a certified HVAC controller.

2. Asset, states and data flow

The asset is one classroom. The principal dynamic state is indoor CO■ concentration; occupancy varies over time and controls generation. Ventilation flow is the controllable input, and a noisy sensor reading represents an observation. Forward flow maps occupancy and ventilation into a predicted state and observation. Reverse/control flow uses the observation to select ventilation, which feeds into the next update. This creates the two-way model/control loop required by a digital-twin concept.

3. Numerical model

The room is approximated as a well-mixed 180 m³ zone with outdoor CO■ fixed at 420 ppm. The simulation uses one-minute steps over six hours. Occupancy is 28 people from minute 20 to 179 and minute 210 to 329. The illustrative generation coefficient is 0.0048 m³ CO■ per person per hour. The update is $C(t+\Delta t)=C(t)+\Delta t/V \times [N \cdot G \cdot 10■–Q(C–Cout)]$, where C is indoor concentration in ppm, N is occupancy, G is generation, Q is ventilation and time is in hours. The model bounds concentration at the outdoor value. Sensor noise is Gaussian with standard deviation 18 ppm. A fixed seed makes the run reproducible.

4. Controller and intelligence layer

The smart policy uses 110 m³/h ventilation while the noisy reading is below 900 ppm, switches to 420 m³/h at or above 900 ppm, and selects 520 m³/h at or above 1050 ppm. Two baselines use constant low or high flow. A second intelligence element flags absolute sensor changes above 120 ppm for review; it does not automatically delete data. A more realistic system would use hysteresis and would distinguish true occupancy transitions from sensor faults.

5. Experiments and outputs

Each policy is run over the same occupancy schedule. Metrics include peak CO■, occupied-period mean, occupied minutes above an illustrative 1000 ppm reference, and mean ventilation flow. The notebook and HTML dashboard show CO■ trajectories, controller decisions, sensor noise and a policy table. The CSV provides minute-level data. The reference line is not asserted to be a universal legal or health threshold; local guidance and other pollutants matter when assessing real indoor air quality.

6. Calibration and identifiability

The notebook sweeps candidate generation coefficients and computes RMSE against a synthetic reference. This demonstrates the mechanics of calibration but is not independent validation because the reference was generated by the same model family. Generation and ventilation can compensate for each other, so one CO■ trace may not uniquely identify both. A field study should collect synchronized CO■, occupancy and ventilation measurements across different operating conditions, fit parameters and report uncertainty intervals.

7. Verification and validation

Automated tests check that CO■ remains at or above the assumed outdoor concentration, repeated runs with a fixed seed reproduce the same trajectory, and high fixed ventilation lowers occupied mean CO■ relative to low fixed ventilation under the selected model. These verify code properties, not real-world accuracy. Validation requires independent sensor data and metrics such as RMSE, mean absolute error, bias and peak-timing error across multiple days.

8. Results and interpretation

The scorecard and plots are generated directly from the deterministic simulation. The high-flow baseline is expected to dilute CO■ more effectively, while the low-flow baseline uses less ventilation but permits more accumulation. The smart policy attempts to increase flow when readings rise, rather than operating at high flow throughout the entire period. Exact numerical outputs are shown in the dashboard and table; they should be reported from those generated results. Flow is used as a proxy for control effort, not as measured electricity consumption, because fan power curves are not included.

9. Credibility scorecard

Conceptual validity is partial because the room is assumed well mixed. Data validity is limited because schedules and parameters are synthetic. Calibration is a sensitivity demonstration, not field calibration. Verification tests pass for the coded assertions, but validation against an independent physical system is outstanding. Uncertainty includes sensor noise but not a full parameter-uncertainty analysis. Operational safety is not established: thresholds are illustrative and the model omits real equipment limits, humidity, particulates, volatile compounds and spatial airflow. CO■ alone is not a complete air-quality measure.

10. Conclusion and future work

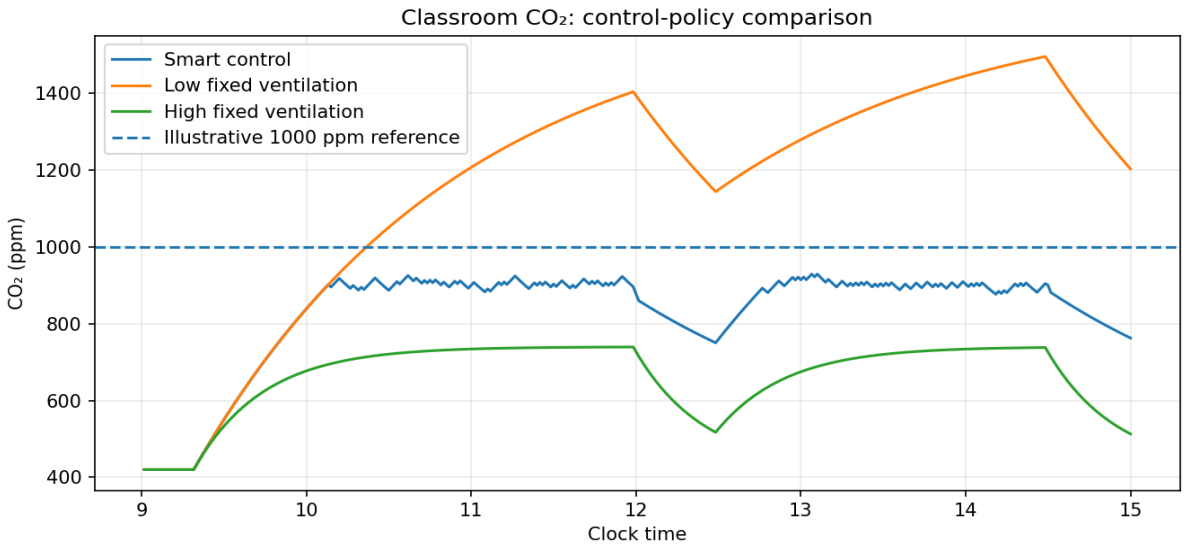
The prototype includes a numerical twin, simulated sensor, controllable ventilation, an interpretable controller, baseline comparison, dashboard, CSV, calibration demonstration, anomaly flags and tests. It demonstrates a reproducible model-to-control loop while making its limitations explicit. Next steps are to collect real classroom data, calibrate and validate against independent periods, propagate parameter uncertainty, use hysteresis to reduce rapid switching, and obtain facilities and safety review before any operational use.

References

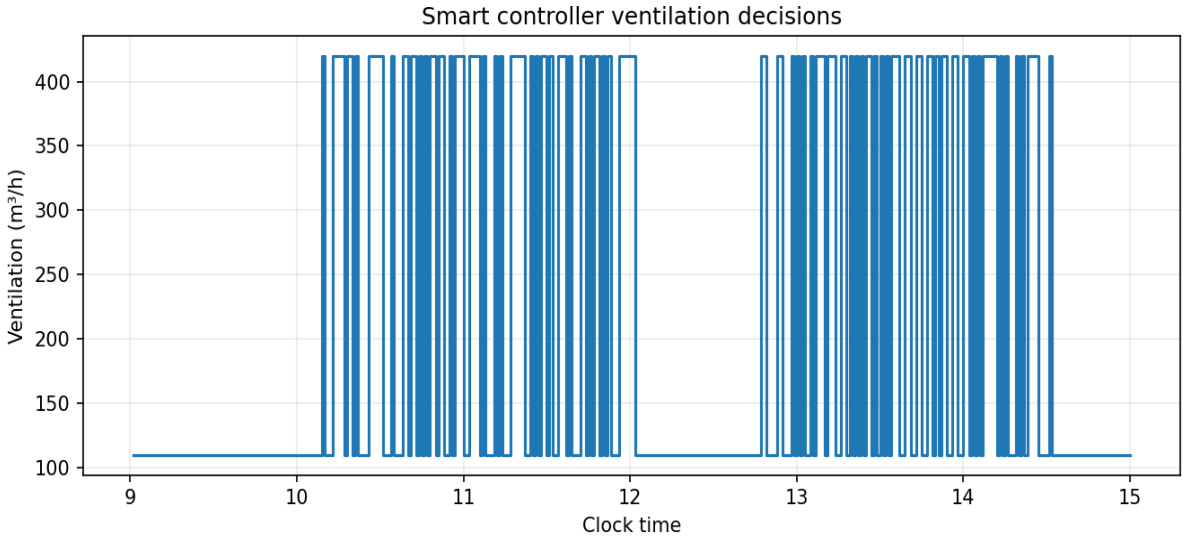
1. utkukose, “Digital Twin with a Credibility Scorecard — Final Project,” digital-twin-veltech-NB-lecture, GitHub: https://github.com/utkukose/digital-twin-veltech-NB-lecture/blob/main/exams/final/README.md (accessed 14 October 2026).
2. Simulation code and figures generated by the accompanying notebook. Parameters not measured are labelled illustrative.

Generated Output Figures

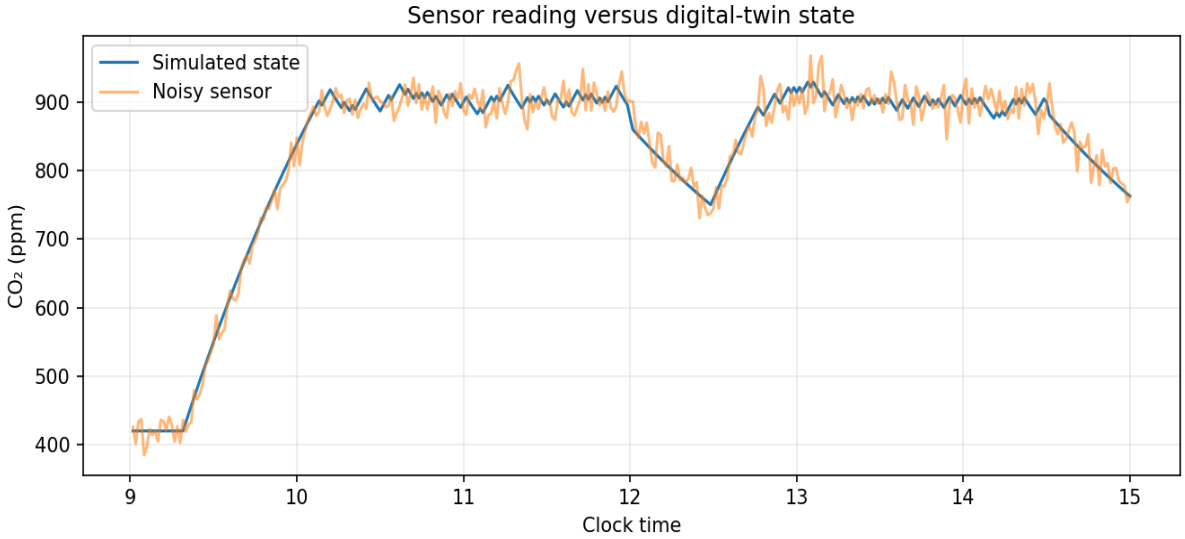
co2 comparison



ventilation control



sensor vs twin



Policy Scorecard

Policy	Peak CO■ (ppm)	Mean occupied CO■ (ppm)	Occupied minutes >1000 ppm	Mean ventilation (m³/h)
smart	929.4	860.4	0.0	211.6
low	1496.1	1178.0	218.0	110.0
high	739.4	692.3	0.0	420.0